

WASP-43b

R-1184

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_220321 · created 2026-07-31T04:50:48+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459659.6864 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.008 +/- 0.051
Transit depth (Rp/Rs)^2	0.007 +/- 0.084 [%]
Orbital Inclination (inc)	77.2 +/- 3.0
Airmass coefficient 1 (a1)	0.885 +/- 0.066
Airmass coefficient 2 (a2)	0.074 +/- 0.078
Scatter in the residuals of the lightcurve fit is	7.37 %
Best Comparison Star	#3 - [592, 367]
Optimal Aperture	6.5
Optimal Annulus	10.28
Transit Duration (day)	0.014 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.008 ± 0.051 vs 0.1594 (deviation 2.97σ)
Duration fit vs published	0.014 d vs 0.0512 d (deviation 2.19σ)
Mid-time O–C	-5.2 min
Depth SNR	0.2
Residual scatter	7.37 %

Light curve

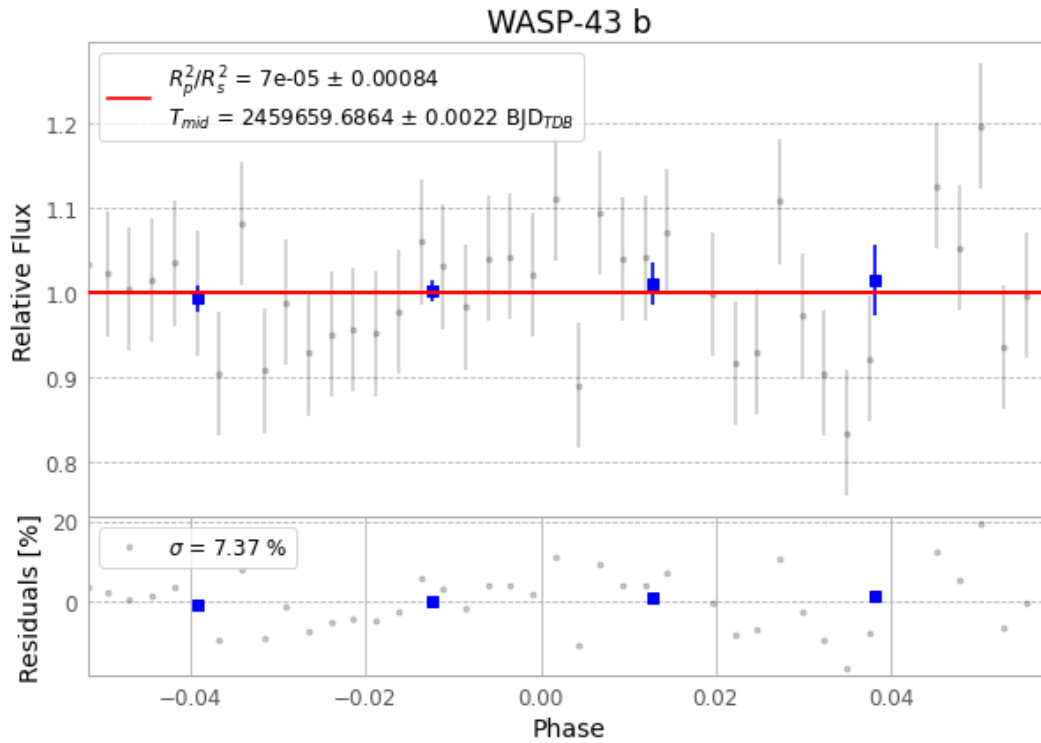


Figure 1 — FinalLightCurve_WASP-43 b_2022-03-20.png (SHA-256 dc9baf339211b65e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 194], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d4cc5446eeced1f0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

55 FITS frames (36 MB), WASP-43220321031844.FITS → WASP-43220321060012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-220321.log (5808b51915ee...), FinalLightCurve_WASP-43 b_2022-03-20.png (dc9baf339211...), FinalLightCurve_WASP-43 b_2022-03-20.csv (c0f5064a1bf8...)

Compute resources

Wall time 280.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 04:50Z.